

Qingcheng Zhao

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🎓 EDUCATION

University of Toronto, Toronto, Canada

Sep. 2025 – Nov. 2030(Expected)

Ph.D. in Computer Science.

Advisors: Prof. Karan Singh

ShanghaiTech University, Shanghai, China

Sep. 2021 – Jun. 2025

Bachelor of Engineering in Computer Science.

Advisors: Prof. Jingyi Yu and Prof. Lan Xu

University of California Berkeley, California, United States of America

Aug. 2023 – Dec. 2023

Exchange Student in College of Engineering, University Of California Berkeley

🔍 RESEARCH INTERESTS

My research interests lie at the intersection of **3D Vision**, **Generative AI**. I am particularly interested in developing generative models for high-fidelity **3D scene reconstruction, rendering, and interaction**, with applications in virtual environments, creative content creation, and embodied AI systems. My work explores both the creation of **3D representations** and the integration of **human-centric priors** to enable context-aware and emotionally responsive interactions in 3D environments. I aim to advance the capabilities of generative AI by bridging the gap between perception and synthesis for real-world and immersive applications.

📖 PUBLICATIONS

- Media2Face: Co-speech Facial Animation Generation With Multi-Modality Guidance **SIGGRAPH 2024**
*Qingcheng Zhao**, Pengyu Long*, Qixuan Zhang, Dafei Qin, Han Liang, Longwen Zhang, Yingliang Zhang, Jingyi Yu, Lan Xu
(Project Page) (Paper)
- DEPR: Depth Guided Single-view Scene Reconstruction with Instance-level Diffusion Priors **ICCV 2025**
Qingcheng Zhao, Xiang Zhang, Zeyuan Chen, Yuan Gao, Zhuowen Tu
- Generating 3D Hair Strands from Images with Arbitrary Styles and Viewpoints **ACM Multimedia 2025**
Pengyu Long, Zijun Zhao, Min Ouyang, *Qingcheng Zhao*, Qixuan Zhang, Wei Yang, Lan Xu, Jingyi Yu

🏢 RESEARCH EXPERIENCE

ShanghaiTech University

Mar. 2022 - Present

Research Assistant. Advised by Prof. Jingyi Yu and Prof. Lan Xu

Co-speech Facial Animation Generation With Multi-Modality Guidance

SIGGRAPH 2024

- Proposed a diffusion model in latent motion space for co-speech facial animation generation, accepting rich multi-modality guidance.
- Built an efficient variational auto-encoder mapping facial geometry and images to a highly generalized and decoupled expression latent space for expressions and identities.
- Achieved state-of-the-art performance on multiple datasets, outperforming existing methods in terms of both quality and diversity.

University of California San Diego

Jul. 2024 - Present

Visiting Scholar. Advised by Prof. Zhuowen Tu

Depth Guided Single-view Scene Reconstruction with Instance-level Diffusion Priors

Under Review

- Proposed a novel depth-guided framework for single-view scene reconstruction that leverages instance-level diffusion priors

- Introduced depth-guided conditioning with local-global attention to encode shape priors into latent tri-plane diffusion models, enabling effective handling of occlusions and strong generalization to real images despite limited synthetic training data.
- Achieved state-of-the-art performance in single-view scene reconstruction through innovative depth-guided DDIM sampling and layout optimization techniques that ensure high geometric consistency between generated reconstructions and input images.

INDUSTRY EXPERIENCE

Nvidia Corporation Shanghai, China

Feb. 2024 – Jun. 2024

Software Development Engineer (Internship)

- Built a LLM-powered agent for gameplay with human-like behaviors in most popular games, using a text-based game UI descriptor to interact with GPT-4.
- Enhanced the language model with a generalizable visual understanding module to improve the agent's performance in various games.
- Widely deployed in production environment, reduced the human labor and time cost significantly, enabling full automation of the game testing process with minimal configuration.

Deemos Technologies Inc. Shanghai, China

Nov. 2022 – Feb. 2024

Intern Researcher

- Built a real-time 3D interactive avatar system utilizing audio-driven facial expression animation technologies at Global AI developer Conference 2023.
- Built a web application for ChatAvatar project based on DreamFace, which can generate 3D avatars from a single image or text prompt.

ACTIVITIES

GeekPie Association

Sep. 2021 - Sep. 2024

President of GeekPie Association

- Led ShanghaiTech's largest developer community, fostering innovation among 200+ members through technical workshops, competitions, and open-source initiatives. Developed  CourseBench, a highly regarded web application that allow students to comment and discuss about courses.
- Organized impactful activities such as SI100+ Python and AI Guide, HPC Competition Training, and the GeekPie Puzzle Challenge; Collaborated with industry leaders (e.g., AMD, Jump Trading) to host seminars and promote cutting-edge technologies like AI and quantitative trading.

CS100: Introduction to Programming

2023, 2024

Senior Teaching Assistant

- Give office hours and recitation classes; Assist with homework assignments and corrections.
- Won the SIST Outstanding Teaching Assistant Award.

HONORS AND AWARDS

<i>ShanghaiTech International Exchange Program First-Class Scholarship</i> , ShanghaiTech University	2024
<i>Outstanding Teaching Assistant</i> , SIST, ShanghaiTech University	2023
<i>Merit Student(top 2%)</i> , ShanghaiTech University	2022
<i>Outstanding Officer</i> , ShanghaiTech Student Union	2022
<i>Bronze Medal</i> , Award on The 2021 China Collegiate Programming Contest, Harbin Site	2021
<i>One-hundred Fourth Place</i> , Award on The 2021 ICPC Asia-East Continent Final Contest	2021

PATENTS

- **Image processing method and device, electronic equipment and storage medium** **CN116017167A**
Inventors: Kuang Cong, Hongyang Lin, **Qingcheng Zhao**, Lan Xu, Yichi Zhang, Hong Wu
 Developed a novel image processing method leveraging layered lighting systems to enhance scene realism in virtual environments. This method integrates virtual lighting systems for high fidelity rendering.

⚙️ SKILLS

- Programming Languages: Python > C/C++ >= Javascript == Typescript > Ruby > MATLAB
- Tools: PyTorch, Blender, OpenCV, Git, L^AT_EX, Docker, Vue, React, FastAPI, Node.js, Rancher, Kubernetes,